

Tribhuvan University
Institute of Science and Technology
2077



Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 109)
(Introduction to Information Technology)
(NEW COURSE)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Section A

Attempt any two questions:

(2×10=20)

1. ✓ Define computer software. Differentiate system software with application software. Explain different types system software in detail. (1 + 2 + 7)
2. ✓ What are the benefits of using computer network? Explain different types of computer network in detail. What is network topology? (3 + 6 + 1)
3. What is database? What are the benefits of storing data in databases? Explain database system architectures in detail. (1 + 3 + 6)

Section B

Attempt any eight questions.

(8 × 5 = 40)

4. ✓ What are different characteristics of a computer? Differentiate digital computer with analog computer. (3 + 2)
5. ✓ What is primary memory? Compare primary memory with secondary memory. (2 + 3)
6. ✓ Define memory hierarchy. Explain different types of ROM in detail. (1 + 4)
7. ✓ Explain any two input devices in detail. (5)
8. Convert $(110101.101)_2$ to decimal. (5)
9. ✓ Define IP address. Why do we need this address? Compare IPv4 address with IPv6 address. (1+ 2+2)
10. ✓ What are different elements of multimedia? Explain. (5)
11. ✓ What is cryptography? How does cryptography provide security to our data? (2 + 3)
12. Write short notes on: (2 × 2.5 = 5)
 - a. ✓ Central processing unit
 - b. ✓ IoT